

Primary Source Reading 1: The Slave Boy and the Diagonal

Optima Academy Online | Philosophy & Ethics | High School Honors (9–12) From Plato's Meno, translated by Benjamin Jowett Read before Lesson 4a: The Geometry of the Soul

Before you read: Socrates has claimed that all learning is really recollection — the soul recovering knowledge it already possesses. To test this, he questions one of Meno's slaves, a boy with no education in mathematics, about a problem in geometry: given a square, how do you construct a new square with exactly double its area? Socrates asks only questions. Watch for the moment the boy is confident and wrong, the moment he realizes he doesn't know, and the moment he works out the answer for himself.

A Confident Wrong Answer

SOCRATES: And might there not be another square twice as large as this, and having, like this, the lines equal?

BOY: Yes.

SOCRATES: And of how many feet will that be?

BOY: Of eight feet.

SOCRATES: And now try and tell me the length of the line which forms the side of that double square: this is two feet — what will that be?

BOY: Clearly, Socrates, it will be double.

SOCRATES: (*to Meno*) Do you observe that I am not teaching the boy anything, but only asking him questions; and now he fancies that he knows how long a line is necessary in order to produce a figure of eight square feet — does he not?

MENO: Yes.

SOCRATES: And does he really know?

MENO: Certainly not.

SOCRATES: He only guesses that because the square is double, the line is double.
(82e–83a)

[Socrates leads the boy to test his guess. If the line is doubled, the boy discovers, the resulting square is not double the area but four times as large — sixteen feet, not eight. The boy then tries three feet, which gives nine — still not eight. He is now stuck.]

The Boy Discovers His Ignorance

SOCRATES: But from what line? — tell me exactly; and if you would rather not reckon, try and show me the line.

BOY: Indeed, Socrates, I do not know.

SOCRATES: Do you see, Meno, what advances he has made in his power of recollection? He did not know at first, and he does not know now, what is the side of a figure of eight feet; but then he thought that he knew, and answered confidently as if he knew, and had no difficulty; now he has a difficulty, and neither knows nor fancies that he knows.

MENO: True.

SOCRATES: Is he not better off in knowing his ignorance?

MENO: I think that he is.

SOCRATES: If we have made him doubt, and given him the torpedo's shock, have we done him any harm?

MENO: I think not.

SOCRATES: We have certainly, as would seem, assisted him in some degree to the discovery of the truth; and now he will wish to remedy his ignorance, but then he would have been ready to tell all the world again and again that the double space should have a double side. (84a-c)

[Socrates begins again, this time drawing a figure of four equal squares and asking the boy to find the diagonal of one of them. Step by step, without being told the answer, the boy works out that a square built on the diagonal of the four-foot square has an area of exactly eight feet.]

The Diagonal

SOCRATES: And this space is of how many feet?

BOY: Of eight feet.

SOCRATES: And from what line do you get this figure?

BOY: From this.

SOCRATES: That is, from the line which extends from corner to corner of the figure of four feet?

BOY: Yes.

SOCRATES: And that is the line which the learned call the diagonal. And if this is the proper name, then you, Meno's slave, are prepared to affirm that the double space is the square of the diagonal?

BOY: Certainly, Socrates.

SOCRATES: What do you say of him, Meno? Were not all these answers given out of his own head?

MENO: Yes, they were all his own. (85b-c)

What the Boy's Own Answers Prove

SOCRATES: And yet, as we were just now saying, he did not know?

MENO: True.

SOCRATES: But still he had in him those notions of his — had he not?

MENO: Yes.

SOCRATES: Then he who does not know may still have true notions of that which he does not know?

MENO: He has.

SOCRATES: And at present these notions have just been stirred up in him, as in a dream; but if he were frequently asked the same questions, in different forms, he would know as well as any one at last?

MENO: I dare say.

SOCRATES: Without any one teaching him he will recover his knowledge for himself, if he is only asked questions?

MENO: Yes.

SOCRATES: And this spontaneous recovery of knowledge in him is recollection? (85c–d)

After you read: Socrates will go on to argue that if the boy's knowledge was in him without anyone teaching it to him in this life, it must have been there before this life — evidence, he claims, for the soul's immortality. He is careful to add that he holds this larger metaphysical claim less confidently than the plainer point it rests on: that the boy's understanding was real, self-generated, and drawn out rather than poured in. Whatever one makes of the soul's prior existence, something in this demonstration deserves an explanation.